

Community Audio Broadcast — Step-by-Step Build Guide

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From an empty bench to a 500-household institution, in ten stages — with what to buy, from where, and in what order.

How to use this guide

Ten stages, roughly twelve months, from an empty bench to one institution serving 500 households. Each stage has a shopping list, a set of steps, and a gate that must pass before the next stage's money is spent.

Spend about ₹2,000 and one week before committing to anything else. Stage 1's decode test decides whether a ₹300 box is possible at all. If a cheap ESP32 cannot decode the stream cleanly, the entire household cost model changes, and it is far better to know that in week one than after a 150-unit order.

The ten stages

Stage	What it produces	Spend	Duration	Gate
0	Decisions, accounts, registrations started	₹7,000	Week 0, then runs in background	DLT application submitted
1	Bench prototype; codec decision settled	₹8,000	2 weeks	Clean audio from a cheap board for 1 hour
2	Mic → internet → phone, end to end	₹2,000/mo	2 weeks	Live audio on a phone at ≤ 8 s
3	Box prototype that plays unattended	₹6,000	3 weeks	Box plays for 7 days untouched
4	Encoder appliance at the institution	₹42,000	3 weeks	Broadcast from their own mixer
5	Firmware: provisioning, OTA, failure handling	₹0	6 weeks	Volunteer installs a box in 3 min
6	Scheduler and prayer timetable	₹0	4 weeks	7 days of slots open unattended

Stage	What it produces	Spend	Duration	Gate
7	25-box field pilot	₹18,000	6 weeks	90% of boxes play every slot for 4 weeks
8	Voice tier live	₹20,000	4 weeks, parallel	85% of test numbers answered or retried
9	150-unit production batch	₹1,70,000	10 weeks	Units pass end-of-line test at $\geq 97\%$
10	Full 500-household pilot	₹30,000	12 weeks	26 weeks unbroken, $\geq 90\%$ reach

Two budgets, kept separate

	What it covers	Total
Per-institution cost	Encoder kit, boxes, enrolment. Recurs at every new institution	$\approx$ ₹1.46 lakh
One-time company cost	PCB design, test jig, certification, tooling, prototypes	$\approx$ ₹1.4 lakh

The second never repeats. Keep it out of the institution's quote, or the first deployment looks twice as expensive as it is.

Critical path

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flowchart LR
  A[Codec test<br/>week 1] --> B[Box prototype]
  B --> C[Firmware + OTA]
  C --> D[25-box pilot]
  D --> E[PCB + batch<br/>10 weeks]
  E --> F[500-household<br/>pilot]
  G[DLT registration<br/>6-8 weeks] --> H[Voice tier]
  H --> F
  I[WPC/BIS cert<br/>8 weeks] --> E

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Three things run on their own clock and must be started early regardless of where the build has reached: **DLT registration** (6–8 weeks), **WPC and BIS certification** (8 weeks), and the **institution's committee approval** (2–4 months). Start all three in Stage 0 or they will each stall the project later.

Conventions used here

Prices are indicative Indian retail at September 2026 and should be re-quoted before ordering. Part numbers are the common generic ones — equivalents from any reputable supplier are fine unless a step says otherwise. Anything marked $\triangle$ is a decision that is expensive to reverse.

Stage 0 — before you buy anything

Week 0. Nothing here needs hardware, and three of these items take two months to clear, so they start now even though they are not needed until Stage 8.

Step 0.1 — Find the pilot institution

Everything else is speculative until one institution has said yes in principle. Requirements: 300–600 households, an active loudspeaker with a daily rhythm, a committee that meets at least monthly, a head who will personally champion it, and a fixed broadband line already on site or available.

Ask for three things in the first meeting: permission to tap the PA mixer for testing, a copy of their printed prayer or service timetable, and their household count. The timetable is needed in Stage 6 and is often surprisingly hard to get later.

Step 0.2 — Start DLT registration $\triangle$

Takes 6–8 weeks and blocks the entire voice tier. Register as a telemarketer on any of the operator DLT portals (Jio, Airtel, VI and BSNL each run one; registration on one propagates).

Item	Cost	Note
Telemarketer registration	₹5,900 incl. GST, one-time	Needs entity PAN, GST, incorporation documents
CLI / header registration	₹590 per number	Register the header early; it is what households see
Annual renewal	₹1,180–2,360	Budget it

Decide first: does the platform register as the sender, or each institution? Registering centrally is far simpler operationally; each institution registering is cleaner legally. For the pilot, register centrally and get a telecom lawyer's view before institution ten.

Step 0.3 — Entity, accounts and legal

Item	Why now
Company incorporation, PAN, GST	DLT registration needs them
Bank account	Component suppliers want business accounts for GST invoices
Cloud account with an Indian region	Mumbai or Hyderabad region for latency and data residency
Domain and email	Certification paperwork and supplier accounts need it
Written consent wording, reviewed against the DPDP Act	Needed before a single household is enrolled
Content governance rules, agreed with the institution	Needed before the first broadcast

Step 0.4 — Settle the design decisions that are expensive to reverse ⚠

Decision	Recommendation for the pilot	Why it is hard to reverse
Codec	Decide in Stage 1 by test, not now	Determines the SoC, which determines the BOM
SoC family	ESP32-S3 with PSRAM for the prototype; fall back to C3 only if the test says it works	Changing after PCB design costs a respin
Audio output path	MAX98357A (I2S DAC and 3 W amp in one chip)	Halves the audio BOM and the board area versus separate DAC and amp
Stream protocol	Icecast-compatible HTTP for boxes	The simplest possible client
Where the schedule lives	Server-side, pushed as play commands	A box that decides for itself when to make noise at 4:45 am is the worse failure

Step 0.5 — Set up the working environment

Tool	Purpose
ESP-IDF v5.x	Firmware build toolchain
A Linux machine or VM	Icecast, encoder testing, build server
Git repository, private	Firmware, backend, and the BOM as a versioned file
A spreadsheet of the BOM with supplier links and dates	This becomes the purchase order at Stage 9

Stage 0 gate

- ☐ Pilot institution identified and agreed in principle
- ☐ Their printed timetable and household count in hand
- ☐ DLT application submitted
- ☐ Entity and cloud accounts live
- ☐ Consent wording drafted

Spend at this stage: ₹7,000, almost all of it DLT registration.

Stage 1 — bench prototype

Two weeks. This stage exists to answer one question: **can a board costing under ₹100 at volume decode a compressed audio stream over Wi-Fi, cleanly, for an hour?** Everything downstream depends on the answer.

Shopping list 1 — bench parts

Buy two of each active component. Things get destroyed on a bench, and waiting a week for a replacement costs more than the spare.

#	Item	Qty	₹ each	₹ total	Source
1.1	ESP32-S3-DevKitC-1-N16R8 , 16 MB flash + 8 MB PSRAM	2	900	1,800	Robu.in, Graylogix, Amazon.in
1.2	ESP32-C3-DevKitM-1, no PSRAM — the cost-target board	2	650	1,300	Robu.in, Robokits
1.3	MAX98357A I2S 3 W class-D amp module	4	89–250	600	Robokits ₹89, Robu, KTron

#	Item	Qty	₹ each	₹ total	Source
1.4	Speaker, 66 mm, 3 W, 4 Ω, full range	4	110	440	Robu, local market
1.5	5 V 2 A USB adaptor, BIS-marked	3	160	480	Amazon.in
1.6	Breadboard, jumpers, USB-C cables, headers	1 set	800	800	Robu
1.7	10 kΩ linear potentiometer with knob	4	30	120	Local market
1.8	USB inline power meter	1	700	700	Amazon.in
1.9	DS3231 RTC module, for Stage 4	2	160	320	Robu
1.10	Ground-loop isolation transformer, 3.5 mm	2	350	700	Amazon.in
1.11	Soldering iron, solder, flux, side cutters	1 set	1,200	1,200	Local market
	Total			₹8,460	

Why MAX98357A rather than a separate DAC and amplifier. It combines the I2S DAC and a 3 W class-D amplifier in one chip. It halves the audio bill of materials, removes a board section, and needs no analogue filtering. At ₹89 for a module on a bench and roughly ₹65 for the bare chip at volume, it is cheaper than the PCM5102 plus PAM8403 pair it replaces.

Step 1.1 — Wire it up

MAX98357A to the ESP32: BCLK, LRC and DIN to three I2S pins, VIN to 5 V, GND to GND, and the gain pin left floating for 9 dB. Speaker to the screw terminals. Volume is handled in software for now; the physical potentiometer comes in Stage 3.

Step 1.2 — Stand up a test stream

On any Linux machine or a cheap VPS: install Icecast2, set a source password, and push a continuous stream into it with ffmpeg from a recorded adhan or sermon file. The point is a real, endless, network-delivered stream — not a file on an SD card.

Encode three versions of the same audio to test against: Opus 32 kbit/s, MP3 64 kbit/s, and AAC 48 kbit/s.

Step 1.3 — The decode test [△](#)

This is the stage's whole purpose. Use a maintained ESP32 audio library rather than writing a decoder.

Test	Board	Codec	Pass condition
A	ESP32-S3 with PSRAM	Opus 32 kbit/s	60 min, zero audible dropouts

Test	Board	Codec	Pass condition
B	ESP32-C3, no PSRAM	Opus 32 kbit/s	60 min, zero audible dropouts
C	ESP32-C3	MP3 64 kbit/s	60 min, zero audible dropouts
D	Either	Any passing codec	Survives Wi-Fi drop and auto-reconnects

Run each for a full hour with the Wi-Fi at about -70 dBm, not next to the router. Log heap free, buffer underruns and reconnects.

Step 1.4 — Read the result

Outcome	Decision	Cost impact
B passes	ESP32-C3 + Opus. The ₹285 BOM holds	Baseline
B fails, C passes	ESP32-C3 + MP3 at 64 kbit/s	BOM holds; bandwidth doubles to $\approx$ ₹30,000/yr per institution
B and C fail, A passes	ESP32-S3 with PSRAM	+₹25–40 per box. BOM goes to $\approx$ ₹320; household price holds at ₹300 and the institution absorbs the gap
All fail	Stop and reconsider the endpoint	Re-plan before spending further

The realistic expectation is that MP3 decodes comfortably on a C3 and Opus is tight without PSRAM. Whichever way it lands, **record the measured numbers in the BOM spreadsheet** — this decision gets questioned repeatedly later and memory of it will not survive.

Step 1.5 — Measure power

With the inline meter, record draw in three states: idle connected, playing at normal volume, and playing at maximum. The design target is under 0.5 W idle and under 2 W playing. If idle exceeds 1 W, investigate Wi-Fi modem sleep before going further — a box that is visible on a household's electricity bill gets unplugged.

Stage 1 gate

- ☐ A board plays a network stream for 60 minutes with no audible dropout

- ☐ It recovers automatically from a Wi-Fi drop
- ☐ Idle power measured and under 1 W
- ☐ Codec and SoC decision written down with the measurements behind it
- ☐ BOM spreadsheet updated with actual prices paid

Spend at this stage: ₹8,460. Cumulative: ₹15,460.

Stage 2 — microphone to phone, end to end

Two weeks, almost no hardware. At the end of this stage someone speaks into a microphone and a phone across the room plays it a few seconds later. That demonstration is also the sales demo from the marketing plan, so build it to be carried.

Shopping list 2 — the capture end

#	Item	Qty	₹	Source
2.1	USB audio interface with line in — Behringer UCA202/UCA222 or equivalent	1	3,000	Amazon.in, local music shops
2.2	Dynamic microphone + XLR-to-3.5 mm lead, for bench testing	1	1,500	Local music shop
2.3	Cloud VPS in an Indian region, 2 vCPU / 4 GB	—	750/month	Any Indian-region provider
2.4	Domain name and TLS certificate	—	900/year	TLS free via Let's Encrypt
	Total, first month		₹6,150	

Step 2.1 — Stand up the relay

Icecast2 on the VPS behind nginx with TLS. Configure one mount point per institution, one source password, and a listener limit. Test with a browser and with VLC before involving any hardware.

Set the burst size deliberately: Icecast's burst-on-connect sends a chunk of buffered audio to each new listener so playback starts instantly. Too large adds latency; too small causes a stutter on join. Start at about one second of audio and tune it in Stage 7 against real field measurements.

Step 2.2 — Build the encoder pipeline

On a laptop for now, moving to the appliance in Stage 4. The chain is: capture from the USB interface, resample to 16 kHz, apply automatic gain control, gate, limit, encode, and push to the Icecast mount.

Element	Setting	Why
Capture	48 kHz stereo from the interface	What the hardware gives
Downmix and resample	Mono, 16 kHz	Speech has nothing above 8 kHz
Automatic gain control	12 dB range, 200 ms attack, 2 s release	Speakers vary enormously in mic distance
Noise gate	−45 dBFS threshold, 30 s hang	This is what drives automatic slot close in Stage 6
Limiter	−3 dBFS ceiling	Stops clipping when someone gets close to the mic
Encoder	Whichever codec Stage 1 chose, constant bitrate	Constant bitrate keeps the box's buffer maths simple
Reconnect	Automatic, exponential backoff	The source must never need a human to restart it

Step 2.3 — Build the listener page

A single HTML page: institution name, one large play button, and a status line. No account, no menu, no settings. It should work on a five-year-old Android phone in a slow browser, which rules out a heavy framework.

Token-gate the stream URL: the page carries a household token, exchanges it for a short-lived stream URL, and plays that. Get this in from the start — retrofitting authorisation onto a public mount is the kind of thing that never gets done.

Step 2.4 — Measure latency properly

Play a sharp click into the microphone and record both the source and the phone output on a second device. The gap between the two clicks is the real glass-to-ear figure. Do not trust the player's own reported buffer.

Target	Value
Glass-to-ear, phone on Wi-Fi	≤ 8 s
Glass-to-ear, phone on 4G	≤ 10 s
Spread between two phones side by side	≤ 3 s

If latency is above 10 s, the cause is almost always the client buffer or Icecast's burst size, not the network. Reduce in that order.

Step 2.5 — Build the demo kit

Pack it into one hard case: laptop, USB interface, isolation transformer, mic lead with a 3.5 mm tap, a mobile hotspot, and two bench boxes from Stage 1. It has to work in a building with no usable Wi-Fi, which is most of them, so the hotspot is not optional.

Stage 2 gate

- ☐ Speak into the mic, hear it on a phone across the room within 8 seconds
- ☐ Stream survives unplugging and replugging the network for 60 seconds
- ☐ Stream URL is token-gated, and an expired token is refused
- ☐ Latency measured with the click test and recorded
- ☐ Demo kit packed and tested in a building that is not the office

Spend at this stage: ₹6,150. Cumulative: ₹21,610.

Stage 3 — box prototype

Three weeks. Turn the breadboard into five enclosed units that look and behave like the real thing, and leave one running for a week without touching it.

Shopping list 3 — five prototype boxes

#	Item	Qty	₹ each	₹ total	Source
3.1	ESP32 devkit, whichever Stage 1 chose	5	650–900	3,900	Robu.in
3.2	MAX98357A module	5	89	445	Robokits
3.3	Speaker, 66 mm 3 W 4 Ω	5	110	550	Robu, local market
3.4	ABS project enclosure, $\sim 110 \times 80 \times 45$ mm, vented	5	90	450	Local market, Amazon.in

#	Item	Qty	₹ each	₹ total	Source
3.5	5 V 1 A BIS-marked adaptor with DC barrel	5	150	750	Amazon.in
3.6	10 kΩ potentiometer with detented off position	5	45	225	Local market
3.7	Bicolour LED and bezel, tactile button	5 sets	40	200	Local market
3.8	Perfboard, DC jack, wire, standoffs	5 sets	120	600	Local market
	Total			₹7,120	

Five units, not one: two go to the field in Stage 7 prep, one lives on a shelf running continuously, one gets taken apart repeatedly, and one goes in the demo kit.

Step 3.1 — Assemble

Perfboard, not a PCB. A PCB at this stage freezes a layout that will change three times. The enclosure is off-the-shelf and will be cut by hand — that is fine and it is what Stage 9 replaces.

Get the physical details right even on a prototype, because they are what the household actually experiences: the volume knob must have a detented off position that is visibly off, the LED must be visible from across a room but not bright enough to annoy at night, and the speaker must face the room rather than a wall.

Step 3.2 — Wire the volume control

The potentiometer is read by the ESP32 as an analogue input and applied as a digital gain, rather than sitting in the analogue signal path. That keeps the knob position readable in telemetry — which is how volume-at-zero becomes an early churn signal — and avoids a scratchy pot degrading the audio.

Step 3.3 — Define the LED language

State	LED	Pairs with
Ready, connected	Steady green	—
Playing	Slow green pulse	Audio
No Wi-Fi	Steady amber	—
Provisioning mode	Fast amber blink	—
Fault	Red	—

Colour is always paired with a blink pattern so the box reads correctly for colour-blind users, and so a photograph sent over WhatsApp by a volunteer is diagnosable.

Step 3.4 — The seven-day soak

One box, plugged in, on normal household Wi-Fi, untouched for seven days. Play a stream into it several times a day, matching the eventual prayer slot pattern. Log every reconnect, every underrun and the free heap.

Pass condition	Value
Missed play commands	0
Unexplained reboots	0
Heap trend over 7 days	Flat, no leak
Reconnects after Wi-Fi interruption	100% recovered without intervention
Audible glitches	≤ 1 per playing hour

A memory leak that takes five days to surface is exactly the fault class that would otherwise be discovered at 400 units in the field. This week is cheap.

Step 3.5 — The listening test

Play a recorded adhan through the prototype to four or five people from the actual community — not engineers, and not in the office. Adhan is melodic rather than plain speech, and it is the most demanding thing this box will ever play.

Ask one question: is this acceptable to listen to every day? If the answer is no, the fix is usually the speaker or the enclosure porting rather than the codec. A better 66 mm driver costs ₹40 more and is the single highest-return upgrade available.

Stage 3 gate

- ☐ Five enclosed units built and working
- ☐ Seven-day soak passed with no leak and no missed commands
- ☐ Community listening test passed on adhan, not just speech
- ☐ LED states finalised and documented
- ☐ Power draw measured in the enclosure, under 0.5 W idle

Spend at this stage: ₹7,120. Cumulative: ₹28,730.

Stage 4 — encoder appliance at the institution

Three weeks. This is the first hardware that leaves the bench and the first time the institution's own microphone is in the chain.

Shopping list 4 — the institution kit

#	Item	Qty	₹	Source
4.1	Raspberry Pi 5, 4 GB	1	5,800	Robu.in, ThinkRobotics, Amazon.in
4.2	Official 27 W USB-C power supply	1	1,000	Same
4.3	Active cooler	1	700	Same
4.4	Industrial microSD, 64 GB, high-endurance	2	900	Amazon.in — buy two, they are the commonest failure
4.5	DS3231 RTC module with battery	1	160	Robu
4.6	USB audio interface (UCA202 or equivalent)	1	3,000	Amazon.in
4.7	Ground-loop isolation transformer, 1:1	1	400	Amazon.in
4.8	Audio leads: RCA, 3.5 mm, 6.35 mm, adaptors	1 set	800	Local music shop
4.9	4G router, Cat-4, with external antenna	1	4,500	Amazon.in
4.10	4G SIM with a data plan	1	1,200/yr	Any operator

#	Item	Qty	₹	Source
4.11	UPS, 600 VA	1	5,000	Amazon.in, local
4.12	Lockable wall-mount enclosure, vented	1	2,800	Local electrical supplier
4.13	Surge protector, earthing, cable, DIN rail	1 set	2,700	Local electrical supplier
4.14	10-inch Android tablet + locked wall mount	1	12,000	Amazon.in
4.14b	<i>Alternative:</i> illuminated arcade button + GPIO wiring	1	1,500	<i>Robu, local market</i>
	Total, tablet variant		₹40,960	

For the pilot, buy both the tablet and the button. Which one the operator actually prefers is worth ₹1,500 to find out, and it decides the standard kit for every institution after this one.

Step 4.1 — Build and harden the appliance

Install a minimal OS, disable everything unnecessary, move logs to RAM to spare the SD card, enable the hardware watchdog, and set up automatic unattended security updates on a fixed weekly window that never coincides with a prayer slot.

Fit the RTC and confirm the time survives a full power-down. Prayer slots depend absolutely on the clock, and a Pi with no RTC that boots after a long outage has no idea what time it is.

Step 4.2 — Move the encoder pipeline onto it

Port the Stage 2 pipeline, run it as a supervised service that restarts on failure, and add a second-level watchdog that reboots the board if the stream has been stalled for five minutes. Test by killing the process, unplugging the interface, and pulling the network cable mid-stream.

Step 4.3 — Set up network failover

Ethernet primary to the fixed line, 4G router on a second interface. Configure so that ingest failure triggers a switch within 15 seconds. Test by physically unplugging the Ethernet mid-broadcast: the stream should drop for a few seconds and resume, not end the session.

Buy the SIM early and check it actually has signal in the equipment cupboard, which is usually an interior room with thick walls. This is what the external antenna is for.

Step 4.4 — Tap the institution's PA [△](#)

This is the step that touches something the institution depends on, so do it on a weekday with nothing scheduled, and with their own technician present if they have one.

1. Find a spare output on the mixer — record out, tape out, aux send, or a second main out. **Never break the existing signal path to the amplifier.**
2. If there is no spare output, use a 1:1 isolation transformer to split the amplifier's input rather than re-patching the mixer.
3. Insert the ground-loop isolator between the mixer and the USB interface. Hum is the single commonest field fault, and this ₹400 part prevents most of it.
4. Set input gain with the muezzin or the priest speaking normally at their usual distance, not with a test tone.
5. Verify the loudspeakers sound exactly as before, with the appliance both powered and unpowered. Have them confirm it themselves.

Step 4.5 — Commissioning checks

Check	Pass
Loudspeaker audio unchanged, appliance on and off	Confirmed by the institution's own ear
No hum or buzz in the stream	Clean at full gain
Speech level consistent between two different speakers	AGC holding
Stream survives Ethernet unplug	Resumes within 15 s
Stream survives mains cut	UPS holds encoder and router 90 min
Clock survives full power-down	RTC holding
Operator can go live unaided	After one 10-minute session

Stage 4 gate

- ☐ Institution's head has spoken into their own microphone and heard it on a phone
- ☐ All seven commissioning checks passed
- ☐ Operator has gone live once without help

☐ Tablet versus button decision made and recorded

Spend at this stage: ₹40,960. Cumulative: ₹69,690.

Stage 5 — firmware

Six weeks, no new hardware. The longest software stage and the one with the least room for error: a firmware bug that strands devices in people's homes cannot be fixed remotely if the bug is in the update path itself.

Step 5.1 — Partition layout first [△](#)

Set this up before writing features, because changing the partition table later means physically re-flashing every device already built.

Partition	Size	Holds
Bootloader	default	—
NVS	24 KB	Wi-Fi credentials, device identity, volume, config
OTA data	8 KB	Which slot is active
App slot A	1.8 MB	Firmware
App slot B	1.8 MB	Firmware, for updates
Storage	remainder	Chime, certificates

Two app slots is what makes rollback possible. Do not economise here.

Step 5.2 — Build the state machine

Boot, Provision, Connecting, Standby, Playing, Reconnect, Offline. Every transition logged, every state having a defined LED output and a defined timeout. Write this as an explicit state machine rather than nested conditionals — the failure modes in Stage 7 are debugged by reading state transitions from telemetry, and that only works if states are real objects.

Step 5.3 — Standby and the play command

A long-lived authenticated connection to the platform in Wi-Fi modem sleep, waking on a keepalive every 30 seconds. A play command carries the stream URL, the volume for this slot, and the playback delay offset.

The box holds no schedule and makes no timing decision. If the platform says play, it plays.

Step 5.4 — Provisioning portal

The target is three minutes, by a volunteer, with no app on their phone:

1. Powered with no credentials, the box raises an access point named for the institution and its own short code.
2. The volunteer joins from any phone; a captive portal opens on its own.
3. The portal lists nearby networks. Pick the household's, type the password, save.
4. The box joins, registers against its pre-assigned household record using its factory identity, and plays a confirmation chime.
5. The office triggers a test tone from the console. Install verified before the volunteer leaves.

The captive-portal redirect is fiddly across Android and iOS versions. Budget a week for it alone and test on at least six different phones, including an old cheap one, because that is what volunteers carry.

Holding the pair button for five seconds returns the box to provisioning — which is how a Wi-Fi password change is handled, and it will be the commonest field task.

Step 5.5 — OTA with rollback [△](#)

Rule	Reason
Fetch in standby only, never during a slot	Never interrupt an adhan
Never between 22:00 and 06:00 local	A reboot chime at 2 am is unforgivable
Verify signature before writing	The fleet is a public-address system in people's homes
Write to the inactive slot, boot once	—
Must register with the platform within 90 s or roll back permanently	This is the whole safety net
Staged: 1%, then 10%, then the rest, 48 h soak each	Catches what testing did not
Automatic halt if any stage exceeds 2% failure	—

Test the rollback path deliberately: sign and ship a firmware that is broken on purpose to a test device, and confirm it comes back on the old slot and reports the failure. An untested rollback is not a rollback.

Step 5.6 — Failure behaviour

Condition	Behaviour
Wi-Fi lost	Amber, exponential backoff to a 5-minute ceiling, retry forever
Stream lost mid-slot	3 reconnects within 10 s, then silence — never a burst of noise
Platform unreachable at slot time	Silent; the platform's own telemetry gap raises the alert
Power cycled	Back to standby automatically, no household action
Volume at zero	Reported as a distinct state — the earliest churn signal available

Step 5.7 — Telemetry

Per slot: joined or not, join latency, playout offset, buffer underruns, RSSI, volume position, uptime, firmware version. Nothing about the household. No audio — the box has no microphone, which is a hardware fact worth being able to state plainly.

Stage 5 gate

- ☐ A person who has never seen the box provisions it in under 3 minutes
- ☐ Tested on six different phones including an old budget Android
- ☐ A deliberately broken firmware rolls back and reports
- ☐ Box survives 50 power cycles with no corruption
- ☐ Telemetry arriving and readable for all five prototypes

Spend at this stage: ₹0. Cumulative: ₹69,690.

Stage 6 — scheduler and timetable

Four weeks. This turns a broadcast tool into something that runs 35 times a week untouched. It is also the most publicly embarrassing place for a bug: adhan at the wrong time, every day, in front of the whole community.

Step 6.1 — Compute the times

Use a maintained prayer-time library. Do not write the astronomy. Configure per institution:

Setting	What it does
Latitude, longitude, elevation	From the institution's actual location, not the town centre
Calculation method	Muslim World League, Umm al-Qura, Karachi, Egyptian, and others — sets the solar depression angles for Fajr and Isha
Asr madhhab	Standard or Hanafi — shifts Asr by 45 to 90 minutes
Per-slot offset	-60 to +60 minutes per prayer, permanent. This is how local practice gets encoded

Step 6.2 — Validate against their printed timetable [⚠](#)

The institution's own timetable is the source of truth, not the computation. Take the printed timetable collected in Stage 0 and diff a full month against the computed times.

Difference	Action
Consistent offset per prayer	Set the permanent per-slot offset. Normal and expected
Varying difference through the month	Wrong calculation method or madhhab. Try others until it agrees
Still disagrees after trying all methods	Import their timetable wholesale. Do not argue with it

Do not open enrolment until a full month agrees. A wrong Fajr is not a bug report, it is a reputational event.

Step 6.3 — Build the slot model

Slot	Duration	Opens
Five daily adhan	~3 min	Computed + offset
Iqamah	~1 min	Fixed minutes after adhan, optional
Jumu'ah khutbah	30–45 min	Fixed weekly
Weekly announcement	5–20 min	Operator or scheduled

Slot	Duration	Opens
Taraweeh, sehri, iftar	Varies	Ramadan calendar, activated by date range
Urgent notice	Any	Operator, immediate, overrides quiet windows

Support at least 8 slots a day and a seasonal slot set that activates and deactivates on a date range. The church equivalents — Angelus three times daily, daily mass, novena — use identical machinery with a different generator, so build the generator as a pluggable piece from the start.

Step 6.4 — The unattended open and close

T	Event
−60 s	Scheduler arms the encoder
−30 s	Platform pre-warms the fleet; boxes connect and buffer
0	Encoder opens the stream; chime plays
—	Muezzin speaks; gate detects audio
+30 s silence, or +6 min hard cap	Encoder closes; boxes return to standby

The pre-warm is what makes the box feel instant. Without it the first four seconds of every adhan are lost to connection setup, and that is exactly the part people notice.

Step 6.5 — Resilience

- **Local cache.** Push 30 days of resolved slot times to the encoder. A slot opens on local time even with the platform unreachable.
- **Nightly recompute** at 02:00, with the coming week visible on the console so an operator can spot an error before the community hears it.
- **Clock guard.** A device whose clock has drifted more than 30 seconds refuses to open a slot and alerts instead.
- **Missed-slot alert.** Slot opened, no audio across the gate within the grace period, alert the operator and support desk. A missed Fajr must be visible by breakfast.
- **Fallback recording.** Off by default. If the institution enables it, a recording of their own muezzin plays after the grace period. Make this a configuration flag with an audit entry, never a default — whether recorded adhan is acceptable is the institution's decision alone.

Step 6.6 — Per-household filtering

The scheduler emits one slot event; the platform fans it out per household against that household's own policy — which slots, what volume, and whether a quiet window suppresses it. A household with Fajr switched off simply receives no play command. All server-side, so the office can change it in seconds without touching the device.

Stage 6 gate

- ☐ A full month of computed times agrees with the printed timetable
- ☐ Seven consecutive days of slots open and close with nobody present
- ☐ Encoder opens a slot correctly with the platform deliberately unreachable
- ☐ Missed-slot alert fires within 2 minutes of a silent slot
- ☐ A household with two slots disabled receives only the other three

Spend at this stage: ₹0. Cumulative: ₹69,690.

Stage 7 — 25-box field pilot

Six weeks. The first time boxes live in real homes on real Wi-Fi. This stage exists to find out what household Wi-Fi is actually like, which is the assumption doing the most work in the entire design.

Shopping list 5 — 25 hand-built boxes

#	Item	Qty	₹ each	₹ total
7.1	ESP32 devkit or module	27	650	17,550
7.2	MAX98357A module	27	89	2,400
7.3	Speaker, 66 mm 3 W	27	110	2,970
7.4	ABS enclosure, off-the-shelf	27	90	2,430
7.5	5 V 1 A BIS adaptor	27	150	4,050
7.6	Potentiometer, LED, button, perfboard, DC jack, wire	27 sets	200	5,400
7.7	Labels, stickers with support number, household cards	30	25	750
	Subtotal, 27 units			₹35,550
	Per unit, hand-built			₹1,317

Note the unit cost. ₹1,317 hand-built from devkits against a ₹285 target at volume. This is normal and it is the gap that Stage 9's PCB, bare modules and contract assembly close. Do not panic about it, and do not let anyone quote it to a committee.

Twenty-seven, not twenty-five: two spares for immediate swaps, because the first field failure will happen in week one and a week's wait destroys that household's confidence permanently.

Step 7.1 — Choose the 25 households deliberately

Not the 25 most enthusiastic. Pick for spread:

Profile	Count	Why
Strong Wi-Fi, technically confident	5	Baseline — these should be flawless
Ordinary household, mid-range router	10	The real case
Weak signal, box far from the router	4	Finds the reconnect bugs
Elderly living alone	4	The persona the product exists for
Known unreliable power	2	Tests resume-on-power-return

Step 7.2 — Pre-bind before the visit

Create the household records and bind each device identity **at the office, before anyone leaves**. The volunteer should never type a household name or a serial number in a living room. Label each box with its household on the outside.

Step 7.3 — Install

1. Place the box where it will live — the room they actually sit in, not the room with the best signal.
2. Provision via the captive portal. Time it with a stopwatch. Target 3 minutes.
3. Record the RSSI the box reports at its final position. This is the number the whole stage is about.
4. Play a test tone from the office console while still in the house.
5. Set the volume with the household present, at the level they want.
6. Leave the household card: what the lights mean, the support number, how to change slots.
7. Explain one thing only: it will play by itself, you do not need to do anything.

Step 7.4 — Measure for four weeks

Metric	Target	What it tells you
Slot join rate within 20 s	$\geq 95\%$	Whether pre-warm works in the real world
Boxes playing every eligible slot	$\geq 90\%$ over 4 weeks	The gate
Playout spread across the 25, p95	≤ 3 s	Whether neighbours hear an echo
Buffer underruns per playing hour	≤ 2	Household bandwidth reality
RSSI distribution	—	How bad household Wi-Fi really is
Wi-Fi password changes during the month	—	Predicts the ongoing field-visit load
Unprompted household complaints	—	The honest signal
Volume at zero after week 2	≤ 2 boxes	Early disengagement

Step 7.5 — Talk to all 25 households at week four

In person, not by survey. Three questions: does it work, is it too loud or too quiet at any time of day, and have you touched it at all. The third question is the important one — any household that has touched the box is a design failure to understand.

What to do if the gate fails

Symptom	Likely cause	Fix
Boxes dropping off overnight	Router dropping idle clients	Shorten the keepalive interval
Poor join rate at weak RSSI	Antenna or placement	Test an external antenna variant; revise placement guidance
Underruns on some households	Genuinely slow connections	Raise the jitter buffer for those endpoints server-side
Several password changes in a month	Normal household behaviour	Make re-provisioning a 60-second task, and budget the visits
Spread above 3 s	Mixed connection quality	Raise the uniform buffer; a consistent 3 s beats a ragged 1–5 s

Stage 7 gate

- ☐ 90% of boxes played every eligible slot for four consecutive weeks
- ☐ Median install time under 5 minutes
- ☐ Playout spread under 3 s at p95
- ☐ Every household interviewed at week four
- ☐ No household has needed to touch the box

Spend at this stage: ₹35,550. Cumulative: ₹1,05,240.

Stage 8 — voice tier

Four weeks of work, running in parallel from Stage 5 onward. The DLT registration started in Stage 0 should be clearing around now, which is why it was started then.

Shopping list 6 — voice

#	Item	₹	Note
8.1	DLT telemarketer registration	5,900	Started in Stage 0
8.2	CLI header registration	590	Per number
8.3	SIP trunk / aggregator setup	8,000–15,000	Varies; negotiate the CPS limit, not just the rate
8.4	Test credit for dial trials	3,000	
8.5	Media server compute	600/month	Small; it only transcodes
	Total setup	≈ ₹20,000	

Step 8.1 — Choose an aggregator

Rate per pulse matters less than three other things:

Ask about	Why it matters
Calls per second permitted	Determines whether 75 households can be reached in 4 minutes
Concurrent channels	40 needed for 75 households at a 50% answer rate
Whether the registered CLI header displays reliably	An unrecognised number at 11 am does not get answered
Answer-detection quality	Paying for voicemail is pure waste
Whether they support bridging into a live audio source	Not all do. Ask before signing

Market rate is around ₹0.10–0.15 per 30-second pulse at pilot volumes, before GST. Do not optimise this line at the cost of the four above.

Step 8.2 — Build the media bridge

A media server that pulls the live stream, transcodes to G.711, and bridges answered calls into it. Late joiners hear the broadcast live from that moment, not from the start. The transcode adds about 40 ms, which is why the voice tier has a tighter latency target than the box tier rather than a looser one.

Step 8.3 — Build the policy engine [△](#)

Four gates before any number is dialled. Every rejection logged with a reason.

1. **Consent** — a dated, stored consent record exists and has not been withdrawn.
2. **Time band** — local time is inside 09:00–21:00.
3. **Slot eligibility** — this slot type is enabled for the voice tier. Prayer slots never are.
4. **Budget** — the institution's monthly voice cap has not been reached.

The budget gate is a **hard stop**, not a warning. It is what stops a stuck retry loop or a misconfiguration producing a bill nobody authorised. Build it before the first real broadcast, not after the first surprise invoice.

Step 8.4 — Build the dialler

Parameter	Value
Calls per second	5, or whatever the aggregator permits
Concurrent channels	40
Ramp to 95% of list	≤ 4 minutes
Ring timeout	25 s
Retries	2, at +3 min and +8 min
Late join	Allowed

Step 8.5 — Test with 30 numbers before committing

Use 30 consenting volunteers, not the real roster. Measure:

Metric	Target	If it fails
Answer rate, first attempt	$\geq 45\%$	Check the CLI header is displaying; it is usually this
Answer rate after 2 retries	$\geq 85\%$	If still low, the tier's economics do not work — re-plan
Ramp time to 95%	≤ 4 min	Negotiate a higher CPS
Audio quality on a feature phone	Intelligible	Check the transcode chain
Calls outside the time band	0	Fix immediately; this is a compliance failure

Step 8.6 — Get a legal opinion [△](#)

Whether a non-commercial religious notice to consenting members falls inside TCCCPR's definition of commercial communication is genuinely unsettled. The system is built as though it does — full consent, full time-band compliance, full DLT registration — because that posture survives either answer. Put it to a telecom lawyer before the first institution goes live, and keep the written opinion.

Stage 8 gate

- ☐ DLT registration and CLI header live
- ☐ 30-number test: $\geq 85\%$ reached within two retries
- ☐ Zero calls placed outside 09:00–21:00 in testing
- ☐ Budget cap tested by deliberately exceeding it
- ☐ Legal opinion in hand

Spend at this stage: ₹20,000 (of which ₹6,500 was already spent in Stage 0). Cumulative: ₹1,18,740.

Stage 9 — 150-unit production batch

Ten weeks and the largest single spend. This is where the box goes from ₹1,317 hand-built to about ₹285 manufactured, and where a mistake is expensive because it is multiplied by 150.

Step 9.1 — Design the PCB

Bare module rather than a devkit. That change alone takes roughly ₹550 out of the unit cost.

Element	Decision
Module	ESP32 module on a castellated footprint, pre-certified radio
Second source	Lay out for a second module vendor on the same footprint wherever possible
Audio	MAX98357A, bare chip
Power	5 V barrel in, 3.3 V LDO. Mains stays outside the enclosure entirely
Connectors	Speaker JST, potentiometer, LED, button, programming header
Test points	Power, I2S, UART and a boot-strap pad — these are what the end-of-line jig uses
Panelisation	Panelised with fiducials, for machine assembly

Item	₹
PCB design and layout, contract, 2 revisions	45,000
Prototype PCBs, 10 units, fast turnaround	6,000
Prototype assembly by hand	3,000
Production PCBs, 160 panelised	14,000
Subtotal	₹68,000

Build and test 10 prototype boards before ordering 160. A respin costs three weeks, and this is the cheapest possible place to find the error.

Step 9.2 — Certification, in parallel [⚠](#)

Start this alongside the PCB, not after. It takes eight weeks and it gates shipping.

Item	₹	Note
WPC ETA self-declaration	25,000	Straightforward because the radio module is already certified — the whole reason for choosing a module over a bare chip
Accredited test-lab report	45,000	
BIS for the adaptor	0	Buy adaptors that are already BIS-registered. Never make your own
RoHS declarations from suppliers	0	Collect at purchase, not later
E-waste recycler registration	15,000	Not needed at 150 units, but set it up before institution ten
Subtotal	₹85,000	

Step 9.3 — Tooling decision

An injection-moulded enclosure costs about **₹1.8 lakh** and pays back past roughly 4,000 units. **At 150 units, do not tool.** Use an off-the-shelf enclosure, machined or laser-cut for the knob, LED and speaker grille. Revisit at institution ten.

Tooling also freezes the mechanical design permanently, so it is worth having several hundred boxes in the field before committing to a mould.

Step 9.4 — Build the end-of-line test jig

Build this even though 150 units could be tested by hand. It is also how field returns get triaged, and retrofitting identity provisioning into a running line is painful.

The jig should, in one operation: power the board, verify Wi-Fi association, play a tone and measure output on a fixed microphone, check current draw in both states, write the device identity and certificate, and print a label. Roughly **₹25,000** in parts and a week of work.

Step 9.5 — Choose the assembler

At 150 units, a local contract shop doing hand or semi-automatic assembly is right. Ask for: a reference customer at similar volume, whether they will work to your BOM without substituting parts, their position on the second-source module, and turnaround for a 150-unit batch.

Never allow silent substitution. A different-but-equivalent speaker or LDO discovered after 150 units are built is a batch-level problem. Put it in the purchase order in writing.

Step 9.6 — Production BOM at 150 units

Item	₹ each	× 160
ESP32 module, bare	95	15,200
MAX98357A, bare chip	65	10,400
Speaker, 66 mm 3 W	32	5,120
Enclosure, off-the-shelf, machined	42	6,720
5 V 1 A BIS adaptor	52	8,320
PCB, pot, LED, button, passives, wiring	28	4,480
Assembly, flashing, test	22	3,520
Landed per unit	₹336	₹53,760

At 160 units the landed cost is about **₹336**, above the ₹285 target. That is expected — ₹285 needs a 500-unit order. The household price stays at ₹300 and the institution's capital budget absorbs the ₹36 gap on this first batch.

Step 9.7 — Quality control on receipt

Check	Sample	Reject if
End-of-line jig pass	100%	Any unit fails
Audio listening check	20 units	Any audible distortion at normal volume
Current draw, idle and playing	20 units	Idle over 0.6 W
48-hour soak	10 units	Any reboot or reconnect failure
Enclosure fit and finish	20 units	Sharp edges, loose knob, rattle

Target end-of-line pass rate is 97% or better. Below 90%, stop and find the cause before accepting the batch.

Stage 9 costs

	₹
PCB design and boards	68,000
Certification	85,000
Test jig	25,000
160 units landed	53,760
Total	₹2,31,760

Of this, **₹1,78,000 is one-time company cost** that never repeats — PCB design, certification and the jig. Only the ₹53,760 of units is per-institution.

Stage 9 gate

- ☐ 10 prototype boards built and passed before the 160 order
- ☐ WPC and BIS paperwork complete
- ☐ Test jig working and provisioning identities
- ☐ 160 units received, 97%+ pass rate
- ☐ 48-hour soak passed on the sample

Spend at this stage: ₹2,31,760. Cumulative: ₹3,50,500.

Stage 10 — full pilot and second institution

Twelve weeks. Almost no engineering. This stage is about 500 households, and it will feel like a step backwards after the technical stages because the work is now social.

Shopping list 7 — launch

#	Item	₹	Note
10.1	Enrolment materials: posters, desk forms, consent forms, volunteer scripts	9,000	In the local language, printed properly
10.2	Household cards, 500	6,000	Fridge-magnet sized: LED meanings, support number
10.3	Volunteer training, 2 sessions	5,000	

#	Item	₹	Note
10.4	Hosted visits and hospitality budget	5,000	For prospective institutions
10.5	Spares pool, 8 additional boxes	2,700	5% of fleet, held at the institution
	Total	₹27,700	

Step 10.1 — Enrolment campaign

Three to four consecutive weekends, immediately after commissioning, while the head's announcement is still fresh.

Weekend	Activity	Cumulative target
0	The head announces it personally, and says why	—
1	Desk after services. Name, address, numbers, tier, consent. T1 households leave with the link playing	40%
2	Repeat desk. Box home visits begin. T3 test calls placed the same day	60%
3	Repeat desk plus installs. The head names the households still missing	75%
4-8	Volunteer door-to-door for the rest, one visit each	85%

Two rules that matter more than they look: **test every T3 number while the person is still at the desk** — numbers are written down wrong constantly, and a test call next week reaches nobody who can fix it. And **never charge ₹300 at the enrolment desk** — the contribution comes later, separately, as a community contribution rather than a price at the point of sign-up.

Step 10.2 — Run 26 weeks

Metric	Target
Consecutive weeks broadcasting	26 of 26
Reach per broadcast, of enrolled	≥ 90%
Endpoints dark, 30-day rolling	≤ 3%
Volume at zero	≤ 2%

Metric	Target
Households disabling all slots	$\leq 5\%$
Slot open success rate	$\geq 99\%$

The three-miss rule is the single most important operational control: any endpoint that misses three consecutive eligible slots generates a volunteer visit. Silent failure is the thing that kills this product, and this is the mechanism that catches it.

Step 10.3 — Write the runbooks while it is fresh

The second institution is onboarded by someone who did not build the first. That only works if these exist and have been used by someone else:

- ☐ Commissioning runbook, encoder and PA tap
- ☐ Box install and provisioning guide, one page, photographs
- ☐ Fault triage sheet, one page — amber means Wi-Fi, no light means adaptor, green but silent means check the console
- ☐ Enrolment day script and materials pack
- ☐ Timetable validation procedure
- ☐ Supplier list with current prices and lead times
- ☐ Operator one-page card

Step 10.4 — Onboard the second institution

The target is **two days of platform effort**, against roughly three weeks for the first. The test is simple and unforgiving: if the original build team has to attend the second institution's first broadcast, the documentation is not finished.

Step	First institution	Target for the second
Tenant setup and configuration	2 days	30 minutes
Timetable validation	1 month	1 week
Encoder commissioning	1 day on site	2 hours, by a partner
Roster import	1 week	1 day
Box provisioning	In the field	Pre-bound before shipping

Pre-binding is the highest-leverage change available. The second institution should receive a carton of boxes already tied to its household records, so a volunteer with no technical skill can do the rest.

Stage 10 gate

- ☐ 85% of households enrolled
- ☐ 26 consecutive weeks, $\geq 90\%$ reach
- ☐ All seven runbooks written and used by someone other than the author
- ☐ Second institution live without the build team present
- ☐ Household contribution collection started and at $\geq 40\%$ participation

Spend at this stage: ₹27,700. Cumulative: ₹3,78,200.

Consolidated purchase schedule

Everything to buy, when, and what it costs. Total to a live 500-household institution: about **₹3.78 lakh**, of which ₹1.78 lakh is one-time company cost that never repeats.

When	What	₹	Running total
Week 0	DLT registration and CLI header	6,500	6,500
Week 1	Bench parts — boards, amps, speakers, tools	8,460	14,960
Week 3	USB interface, mic, VPS, domain	6,150	21,110
Week 5	5 prototype boxes	7,120	28,230
Week 8	Institution kit — Pi, UPS, router, tablet, cabling	40,960	69,190
Week 20	27 hand-built field-pilot boxes	35,550	1,04,740
Week 22	Voice — aggregator setup and test credit	13,500	1,18,240
Week 26	PCB design and prototype boards	54,000	1,72,240
Week 26	Certification — WPC, test lab, e-waste	85,000	2,57,240
Week 28	End-of-line test jig	25,000	2,82,240
Week 30	Production PCBs, 160 panelised	14,000	2,96,240
Week 32	160 units — components and assembly	53,760	3,50,000
Week 38	Enrolment materials, cards, training, spares	27,700	3,77,700

Split by who bears it

	₹	Recurrs at the next institution?
Institution kit	40,960	Yes
160 boxes	53,760	Yes, scaled to household count
Enrolment and launch	27,700	Yes
Voice setup (aggregator, not DLT)	7,000	Partly
Per-institution subtotal	₹1,29,420	
Prototypes and bench	21,730	No
PCB design and boards	68,000	No
Certification	85,000	No
Test jig	25,000	No
DLT registration	6,500	No
One-time company subtotal	₹2,06,230	

The per-institution figure of about ₹1.29 lakh lines up with the ₹1.46 lakh capital estimate in the business case once contingency is added. **Quote the institution the first number, never the total** — the certification and PCB work is the company's, not theirs.

Monthly running costs, from week 3

Item	₹/month
Cloud VPS and bandwidth	1,250
Media server for voice	600
4G SIM at the institution	100
Voice minutes, weekly tier only	1,580
Notifications	500
Total	₹4,030

Where the money actually goes

At ₹3.78 lakh total, the three largest lines are **certification (₹85,000)**, **PCB design (₹68,000)** and **the 160-unit batch (₹53,760)**. Two of those three are one-time. The instinct is to economise on the batch; the better economy is to get the PCB right first time, because a respin costs ₹20,000 and three weeks of everyone's schedule.

Suppliers and long-lead items

Where to buy what

Category	Suppliers	Notes
ESP32 boards and modules	Robu.in , Robokits, Graylogix, ThinkRobotics, Amazon.in	Robu and Robokits carry the widest Espressif range with GST invoices
Audio modules	Robokits at ₹89, Robu, KTron, Fab.to.Lab	Prices vary 3× between suppliers for the same module — compare before bulk buying
Raspberry Pi	ThinkRobotics , Robu, Silverline, Amazon.in	4 GB at ₹5,500–6,200. Check the seller is an authorised reseller
Bare components at volume	Mouser India, DigiKey India, Element14 India, LCSC	For the 150-unit batch. Long lead times — order early
Speakers, enclosures, passives	Local markets: SP Road Bengaluru, Lamington Road Mumbai, Nehru Place Delhi, Ritchie Street Chennai	Cheapest and fastest for prototype quantities; insist on GST bills
PCB fabrication	Indian fabs for short runs; JLCPCB or PCBWay for cost, with import duty and 2–3 weeks	Get quotes from both
Audio interfaces, UPS, routers, tablets	Amazon.in, local music and IT shops	
Contract assembly	Local EMS shops — find via the PCB fab or an electronics association	Ask for a reference customer at similar volume

Long-lead items — order first

Item	Lead time	Blocks	Start at
DLT registration	6–8 weeks	The entire voice tier	Week 0
WPC ETA and test-lab report	8 weeks	Shipping any box	Week 24 , alongside PCB design
Committee approval at the institution	2–4 months	Everything on site	Week 0
Bare components from international distributors	3–6 weeks	The 160-unit batch	Week 26
PCB fabrication and assembly	3–4 weeks	The batch	Week 28
Contract assembler onboarding	2–3 weeks	The batch	Week 26
Fixed broadband at the institution	2–4 weeks	Commissioning	Week 4

The three in bold are the ones that stall projects. None of them can be accelerated with money, and all three can be started before a single component is bought.

Order-first checklist, week 0

- ☐ DLT telemarketer registration submitted
- ☐ Pilot institution agreed in principle; committee date in the diary
- ☐ Their printed prayer or service timetable collected
- ☐ Fixed broadband ordered at the institution if not already present
- ☐ Entity, GST, bank account and cloud account live
- ☐ Consent wording drafted for DPDP review
- ☐ BOM spreadsheet created, with a column for actual price paid

Buying advice

Always take the GST invoice. Input credit on ₹3.78 lakh is not trivial, and certification paperwork needs traceable purchases.

Buy two of everything active on the bench. A ₹900 spare board beats a week of waiting.

Never substitute silently, and never let an assembler do it. Put it in writing on the purchase order.

Qualify a second source for the ESP32 module before you need it. Module allocation shortages have happened repeatedly and a four-week gap that stops 4,000 installs is far worse than ₹3 more per unit.

Buy adaptors that are already BIS-registered. Never build your own mains supply. It moves safety certification onto your board and it is not worth the ₹20 saved.

Appendix A — 8-week student project track

Scope: what eight weeks actually buys

The main guide runs 38 weeks. A final-year team of four has about eight, alongside other coursework. So this track does **Stages 1, 2, 3 and useful slices of 5 and 6**, and stops there deliberately rather than doing all ten badly.

The deliverable

A working five-box demonstrator: a microphone in one room, five enclosed Parish Boxes in five different real homes, playing live audio on an automatic daily schedule with nobody pressing anything — plus a measurement report that answers the codec question with data.

That is a complete, defensible final-year project. It has a real research question, a hardware build, embedded firmware, a distributed system, and field measurement on somebody else's Wi-Fi rather than the lab's.

What is in

From the main guide	What the team does
Stage 1	Bench prototype and the decode test — the project's research question
Stage 2	Microphone → encoder → stream → phone, with measured latency
Stage 3	Five enclosed boxes, LED states, volume knob, 7-day soak
Stage 5 (part)	State machine, Wi-Fi provisioning portal, telemetry. Not OTA

From the main guide	What the team does
Stage 6 (part)	Prayer-time computation, unattended slot open and close. Not the 30-day encoder cache
Stage 7 (shrunk)	5 boxes in 5 homes for 10 days, instrumented

What is cut, and why

Cut	Reason
PCB design and the 160-unit batch	10 weeks and ₹1.4 lakh. Perfboard proves the same engineering
WPC / BIS certification	8 weeks of paperwork, no learning value, and not needed for a lab demonstrator
DLT registration and the voice tier	6–8 weeks of regulatory lead time entirely outside the team's control
The ₹41,000 institution kit	A laptop or a spare Pi does the encoding. UPS, 4G router and tablet add nothing to the demonstration
OTA with rollback	Genuinely hard to do safely. Attempt it only as a stretch goal, on the bench, never on the field boxes
500-household enrolment	Social work, not engineering, and it needs an institution that has approved a 2–4 month committee process

The research question

An assembly job is not a final-year project. This one is, because of a question nobody has a confident published answer to:

Can an ESP32-C3 — single-core RISC-V, no PSRAM, under ₹100 at volume — decode a compressed audio stream over Wi-Fi with zero audible dropouts for 60 minutes at –70 dBm? And what does the answer cost?

That question has a measurable answer, a clear method, a fallback path (MP3 at 64 kbit/s), and a real consequence: if the answer is no, the household device cost rises by ₹25–40 and a ₹300 ceiling is at risk. Frame the whole report around it.

A second, softer question is worth carrying alongside, because it is where the field data goes:

How bad is real household Wi-Fi, and does RSSI at the box's actual position predict whether a scheduled slot is missed?

Team size and effort

Four students, roughly 10–12 hours a week each. Three students can do it by dropping the field trial to three homes. Two cannot — the firmware and the backend cannot be the same person in eight weeks.

The 8-week plan

Every week has one deliverable and one gate. A week that does not pass its gate is a week to report on honestly, not to hide — a documented negative result is worth more marks than a vague positive one.

```
flowchart LR
    W0[Week 0<br/>order parts] --> W1[Week 1<br/>decode test]
    W1 --> W2[Week 2<br/>mic to phone]
    W2 --> W3[Week 3<br/>5 boxes built]
    W3 --> W4[Week 4<br/>provisioning]
    W4 --> W5[Week 5<br/>scheduler]
    W5 --> W6[Week 6<br/>deploy to homes]
    W6 --> W7[Week 7<br/>10 days of data]
    W7 --> W8[Week 8<br/>analyse + report]
```

Week 0 — before the clock starts

If possible, do this in the week before the project formally begins. **Parts take 5–10 days to arrive and that delay is the commonest reason these projects run out of time.**

- ☐ Order everything on the shopping list below, in one go
- ☐ Assign the four roles and agree who owns which repository
- ☐ Install the toolchain: ESP-IDF or PlatformIO, and a Linux machine or VM for Icecast
- ☐ Create one private Git repo with three folders: `firmware/`, `backend/`, `measurements/`
- ☐ Create the BOM spreadsheet with a column for **actual price paid**, not list price
- ☐ Identify five households willing to host a box for 10 days in week 6

Gate: parts ordered, repo created, five host households identified by name.

Week 1 — the decode test

The project's research question, answered first rather than last. Do not build anything pretty this week.

- Wire one ESP32-S3 and one ESP32-C3 to a MAX98357A on breadboard
- Stand up an Icecast server and push a looping recorded adhan into it in three encodings
- Run tests A-D from Stage 1 for a full hour each, at about -70 dBm, logging free heap, buffer underruns and reconnects
- Record power draw in idle, playing and maximum states

Gate: a table of four test results with numbers, and a written codec + SoC decision. **If nothing passes, that is a publishable result** — report it, take the MP3 fallback, and carry on.

Week 2 — microphone to phone

- Capture from a USB audio interface, resample to 16 kHz mono, AGC, gate, limiter, encode, push to Icecast
- Build the one-button listener web page
- Measure end-to-end latency with the single-recorder method described below
- Measure the spread between two phones side by side

Gate: speak into the mic, hear it on a phone across the room, and state the latency as a measured number with a method — not an estimate.

Week 3 — five boxes

- Move from breadboard to perfboard. Five units, in enclosures, with speaker, pot, LED and DC jack
- Cut the enclosures properly. The boxes go into somebody's living room in three weeks
- Implement the five LED states
- **Start the 7-day soak on one box today.** It runs in the background while week 4 happens

Gate: five working enclosed units; soak box logging.

Week 4 — provisioning and telemetry

- Implement the state machine: Boot, Provision, Connecting, Standby, Playing, Reconnect, Offline

- Build the captive-portal provisioning flow and **test it on at least four different phones**, including one old cheap Android
- Implement telemetry: join success, join latency, RSSI, underruns, volume position, uptime
- Stand up the telemetry backend and a minimal dashboard

Gate: a person outside the team provisions a box in under five minutes, unaided, and it appears on the dashboard.

Week 5 — unattended operation

- Compute prayer times from coordinates, method and madhhab
- Slot open and close: chime, gate detection, close on 30 s silence or a 6-minute cap
- Pre-warm the fleet 30 s before each slot
- Run five slots a day for seven days on the bench with nobody touching anything

Gate: seven consecutive days of scheduled slots opening and closing with no human present, logged.

Week 6 — into real homes

The most valuable week and the one most likely to go wrong. Protect it.

- Pre-bind each device to its household record **before** leaving the lab
- Install five boxes in five homes with genuinely different Wi-Fi conditions — not five hostel rooms on the same access point
- Record RSSI at each box's final position, and time each install with a stopwatch
- Leave a card with the LED meanings and a phone number

Gate: five boxes live in five distinct networks, all reporting telemetry.

Week 7 — collect

Ten days of data across weeks 6 and 7. **Do not change firmware during this window.** A mid-trial firmware change invalidates the dataset and there is no time to rerun it.

- Daily: join rate, playout spread, underruns, dark endpoints
- At the end: interview all five households in person. Ask whether they touched the box

Gate: a clean 10-day dataset with no firmware changes inside it.

Week 8 — analyse and write

- Answer both research questions with plots, not adjectives
- Correlate RSSI against missed slots — this is the most interesting analysis available

- Write the BOM with actual prices paid and extrapolate to 500 and 5,000 units
- Record a 5-minute demonstration video, because live demos fail on lab Wi-Fi
- Write the report and prepare the viva

Gate: report submitted, demo video recorded, repository tagged.

The two weeks that must not slip

Week	Why
Week 1	It decides the chip and the codec. Every later week is built on that answer, and discovering in week 5 that Opus does not fit is unrecoverable
Week 6	The field trial needs 10 clear days before week 8. Slipping the deployment by one week deletes the entire field dataset, which is the part that makes this a project rather than a lab exercise

If the team is running late in week 3 or 4, cut the LED polish, cut the dashboard styling, cut the enclosure finish. **Do not cut the field trial.**

Student shopping list

₹18,600 total, against ₹3.78 lakh for the full programme. The difference is almost entirely certification, PCB tooling and a production batch — none of which a demonstrator needs.

Buy

#	Item	Qty	₹ each	₹ total	Note
S1	ESP32-S3-DevKitC-1-N16R8 (16 MB flash, 8 MB PSRAM)	2	900	1,800	The PSRAM board — test A
S2	ESP32-C3-DevKitM-1 (no PSRAM)	3	650	1,950	The cost-target board — tests B and C
S3	MAX98357A I2S amplifier module	8	89	712	DAC and 3 W class-D in one part. Buy spares
S4	Speaker, 66 mm, 3 W, 4 Ω, full range	8	110	880	
S5	ABS project enclosure, ~110 × 80 × 45 mm	6	90	540	Vented, or drill your own grille
S6	5 V 1 A BIS-marked adaptor, DC barrel	8	150	1,200	Never build your own mains supply

#	Item	Qty	₹ each	₹ total	Note
S7	10 kΩ linear potentiometer with knob	8	45	360	Detented off position if available
S8	Bicolour LED, bezel, tactile button	8 sets	40	320	
S9	Perfboard, DC jacks, headers, hookup wire, standoffs	1 set	1,400	1,400	
S10	Breadboard, jumper wires, USB-C cables	1 set	900	900	
S11	INA219 current sensor module	2	150	300	For sub-watt standby measurement. See below
S12	DS3231 RTC module with battery	2	160	320	Slot timing after a power cut
S13	USB sound card with line/mic in	1	700	700	A ₹3,000 UCA202 is nicer; ₹700 is sufficient
S14	Ground-loop isolation transformer, 3.5 mm	2	350	700	Hum is the commonest audio fault
S15	Cloud VPS, 2 months	1	750/mo	1,500	Indian region. A college server works too
S16	Consumables: solder, flux, heatshrink, tape	1 set	900	900	
S17	Contingency — things get destroyed	—	—	4,000	Do not skip this line
	Total			₹18,482	

Borrow from the department instead of buying

Item	Typical retail	Borrow from
Soldering station, side cutters, tweezers	₹3,500	Any electronics lab
Digital multimeter	₹1,500	Lab
Oscilloscope (useful for I2S debugging)	—	Lab
Dynamic microphone and XLR lead	₹1,500	AV / music society
Laptop or Raspberry Pi as the encoder	₹5,000–60,000	Any team laptop will do
Wi-Fi router for the controlled bench test	₹1,500	Lab or a hostel room
Phone for the latency recording	—	Anyone's

The laptop substitution matters: the main guide's ₹41,000 institution kit exists to survive power cuts and run unattended for years. A demonstrator needs neither, and **a laptop running the encoder pipeline is functionally identical for eight weeks.**

Where to buy

Category	Supplier
ESP32 boards	Robu.in, Robokits, Graylogix, ThinkRobotics, Amazon.in
MAX98357A modules	Robokits (cheapest at ~₹89), Robu, KTron, Fab.to.Lab
Speakers, enclosures, passives, pots	Local markets — SP Road Bengaluru, Lamington Road Mumbai, Nehru Place Delhi, Ritchie Street Chennai
Adaptors, sound card, sensors	Amazon.in

Order everything in one go in week 0. Splitting the order across three weeks is how a two-month project loses two weeks. Prices vary up to 3× between Indian suppliers for the same module — compare before ordering, but do not spend three days saving ₹200.

If the budget is ₹10,000

Drop to three boxes instead of five, buy one S3 and two C3 boards, skip the INA219 (use a USB inline power meter at ₹700 and accept coarser standby figures), and use a college machine instead of a VPS. The project still works; the field trial is just smaller, and the report should say so.

Wiring and pinout reference

The main guide assumes you know this. Here it is explicitly, including the three traps that cost teams a week each.

MAX98357A to ESP32

Module pin	Connect to	Note
VIN	5 V rail	Not 3.3 V — output power depends on it
GND	GND	Common ground with the ESP32

Module pin	Connect to	Note
BCLK	Any free GPIO	I2S bit clock
LRC	Any free GPIO	I2S word select, left-right clock
DIN	Any free GPIO	I2S serial data
GAIN	Leave floating	9 dB. To GND = 12 dB; to GND via 100 k Ω = 15 dB; to VIN = 6 dB
SD	Tie to VIN via 100 k Ω	Shutdown and channel select, not an SD card. See trap 2
Speaker + / -	Speaker terminals	The output floats — never ground either side of a class-D output

A pin map that avoids strapping pins on both boards:

Signal	ESP32-S3	ESP32-C3
I2S BCLK	GPIO5	GPIO4
I2S LRC	GPIO6	GPIO5
I2S DIN	GPIO7	GPIO6
Volume pot, ADC	GPIO1	GPIO0
LED green	GPIO15	GPIO10
LED amber	GPIO16	GPIO7
Pair button	GPIO17	GPIO3

Trap 1 — strapping pins

Some GPIOs are sampled at reset to choose boot mode. Hanging an LED or a button on one stops the board booting, and the symptom is a board that worked on the bench and dies the moment you wire the front panel.

Board	Avoid
ESP32-S3	GPIO0, GPIO3, GPIO45, GPIO46
ESP32-C3	GPIO2, GPIO8, GPIO9
ESP32 classic	GPIO0, GPIO2, GPIO12, GPIO15

Check the datasheet for your exact module rather than trusting a blog pinout diagram.

Trap 2 — the SD pin is not an SD card

On the MAX98357A, SD means shutdown, and it also selects the channel by voltage. Left floating or low, the amplifier is silent and you will spend an afternoon debugging your I2S code for nothing.

SD voltage	Behaviour
Below ~0.16 V	Shutdown, no output
~0.16–0.77 V	Left channel only
~0.77–1.4 V	Right channel only
Above ~1.4 V	(L+R)/2 — use this for a mono stream

Most breakout boards already fit a pull-up. Measure it before assuming.

Trap 3 — ADC2 does not work while Wi-Fi is on

This one is specific to the ESP32 family and it will absolutely catch somebody. **ADC2 is shared with the Wi-Fi radio.** Read the volume potentiometer on an ADC2 pin and it returns garbage, or blocks, whenever Wi-Fi is active — which on this device is always.

Put the pot on an **ADC1** pin. On the ESP32-S3 that is GPIO1–GPIO10; on the ESP32-C3, GPIO0–GPIO4. Also note the ADC is not linear at the rails, so map roughly 100–4000 counts to your 0–100% volume range rather than 0–4095, and apply hysteresis of a few counts or the volume will jitter audibly.

Volume and LED

The pot is read as an ADC input and applied as **digital gain in software**, not placed in the analogue signal path. Two reasons, both worth putting in the report: the knob position becomes readable in telemetry, and a cheap scratchy pot never degrades the audio.

Wire the pot as a divider: one outer leg to 3.3 V, the other to GND, wiper to the ADC pin. Add a 100 nF capacitor from wiper to GND to quieten it.

For the LED, either two GPIOs driving a bicolour LED through 330 Ω resistors, or the onboard addressable RGB LED that most devkits carry. The addressable one is easier to wire and harder to see across a room — for the field boxes, use a real LED on the front panel.

Power wiring

The 5 V adaptor feeds the module's VIN and the devkit's 5 V pin. The devkit's onboard regulator supplies 3.3 V to the SoC. **Do not power the board from USB and the adaptor at the same time** unless the devkit has reverse-protection diodes — many do, some do not, and the failure is expensive.

Decoupling: a 470 μ F electrolytic across the 5 V rail near the amplifier, plus 100 nF close to each IC. Class-D amplifiers draw current in bursts, and without bulk capacitance you get audible clicks on loud passages that look exactly like a software buffer bug.

The software chain

Four pieces: a relay, a source, a client, and a scheduler. Build them in that order — each one is testable before the next exists.

1. The relay (Icecast)

On a VPS or any Linux machine the boxes can reach. On Debian or Ubuntu, install `icecast2` and edit `/etc/icecast2/icecast.xml`. Four things matter:

Setting	Value
<code><source-password></code>	Something long. This is what lets a source publish
<code><listen-socket><port></code>	8000
<code><burst-size></code>	About one second of audio. Too large adds latency; too small stutters on join
<code><hostname></code>	The machine's real IP or name — wrong here and mount URLs break

Check it works by opening `http://<host>:8000/` in a browser before touching any hardware. **Test each layer alone.** A student team that wires the ESP32 first and then debugs "no audio" against an unverified server loses two days.

2. The source (ffmpeg)

A live microphone into the relay. On Linux, find the capture device with `arecord -l`, then:

```
ffmpeg -re -f alsa -i hw:1,0 -ac 1 -ar 16000 \
  -af "highpass=f=80,acompressor=threshold=-20dB:ratio=4,alimiter=limit=0.9" \
  -c:a libmp3lame -b:a 64k -content_type audio/mpeg \
  -f mp3 icecast://source:PASSWORD@HOST:8000/live.mp3
```

The `-af` chain is your AGC, gate and limiter in crude form — good enough for week 2, and worth replacing with a proper chain later if the audio student wants a sub-project. For the Opus variant:

```
ffmpeg -re -f alsa -i hw:1,0 -ac 1 -ar 16000 \
  -c:a libopus -b:a 32k -content_type application/ogg \
  -f ogg icecast://source:PASSWORD@HOST:8000/live.opus
```

For week 1's soak tests, replace `-f alsa -i hw:1,0` with `-stream_loop -1 -i adhan.wav` so a recording loops forever without anyone standing at a microphone.

3. The client (ESP32)

Option	Framework	Use when
ESP32-audioI2S (schreibfaul1)	Arduino / PlatformIO	Start here. <code>connecttohost()</code> takes an Icecast URL and plays it. Working audio in an afternoon
ESP-ADF	ESP-IDF	The production path. HTTP stream element plus decoder elements, far more control, much steeper

Start with the Arduino library to get sound out of a speaker in week 1, then decide whether the firmware student moves to ESP-ADF. Getting audio working early matters more than architectural purity in an 8-week project.

Why MP3 is the easier first target

The main guide specifies Opus at 32 kbit/s, and for the platform that is right — best speech quality per bit, royalty-free. But for a student team it is the harder starting point, and the reason is the container, not the codec:

- Icecast carries Opus inside an **Ogg** container. The ESP32 must demux Ogg pages before it can decode anything.
- MP3 over Icecast is a raw frame stream with no container at all. The decoder just consumes bytes.
- Opus decode also wants more RAM than MP3, which is exactly what a C3 without PSRAM does not have.

Get MP3 at 64 kbit/s working first. It proves the whole chain — network, I2S, amplifier, speaker — in one go. Then attempt Opus as the actual experiment, with a known-good baseline to compare against. A team that starts with Opus-in-Ogg on a C3 can spend three weeks unsure whether the problem is their wiring, their buffer, or the codec.

That comparison is the heart of the report: **MP3 64 kbit/s versus Opus 32 kbit/s, on two boards, measured.**

4. The scheduler

A small Python service on the same machine as the relay. Use a maintained prayer-time library rather than writing the astronomy — the Adhan family has well-tested ports in Python, JavaScript and C. Configure latitude, longitude, calculation method and Asr madhhab, plus a per-slot offset.

What it does	Detail
Compute	Next 7 days of slot times, recomputed nightly
Arm	Start the ffmpeg source at T-60 s
Pre-warm	Tell the boxes to connect at T-30 s
Close	Stop at 30 s of silence or a 6-minute cap
Log	Every slot: opened, audio detected, boxes joined

For the box control channel and telemetry, two workable choices:

Approach	Good for
MQTT (Mosquitto broker)	Fastest to build. Boxes subscribe to a topic, publish telemetry to another. Natural fit for the play/stop command model
HTTP + SQLite (Flask or FastAPI)	More control, easier to query for the report, one less moving part to install

MQTT is the better fit for the design and the faster path. Whichever you choose, **log every telemetry message to a database from day one** — the report's plots come from that table, and you cannot reconstruct week 6 in week 8.

Repository layout

firmware/	ESP32 source, one PlatformIO or IDF project
backend/	relay config, scheduler, telemetry API, dashboard
measurements/	raw logs, notebooks, plots, the BOM spreadsheet
docs/	wiring photos, test procedures, the report

Commit the raw measurement data, not just the plots. An examiner asking to see the numbers behind a graph is a normal question, and "we regenerated it" is not an answer.

Measurement methods

The difference between a build and a project is that a project measures things. Four measurements carry this report, and each has a method that avoids an obvious error.

Latency — use one recorder, not two

The naive method is to record the source on one device and the box output on another and compare timestamps. **Do not do this.** Two devices have two clocks with unknown offset and drift, and the error is easily larger than the thing you are measuring.

The single-recorder method:

1. Put the box and the source loudspeaker in the same room, roughly equidistant from one phone or recorder set to record.
2. Clap once, sharply. The clap reaches the microphone directly and also travels through the whole chain and out of the box.
3. The recording now contains both transients in **one file, on one clock**.
4. Open it in any audio editor, measure the gap between the two onsets, subtract the acoustic flight time (about 3 ms per metre).
5. Repeat 20 times. Report median and the 5th and 95th percentiles — not a single number.

This is accurate to a few milliseconds with no special equipment, and it is the method to describe in the report.

For the spread between endpoints, put two boxes side by side and clap once. Both outputs land in the same recording, and the gap between them is the spread directly.

Standby power — below what a USB meter can see

A USB inline power meter resolves about 10 mA, which at 5 V is 50 mW. You are trying to measure whether standby is 350 mW or 900 mW, so the meter is usable but coarse, and it cannot show you the duty cycle.

Better, and worth the ₹150: an **INA219** current sensor in series with the 5 V rail, reporting over I2C to a second microcontroller or a Raspberry Pi, logging at 10 Hz for an hour.

This matters because Wi-Fi modem sleep is not a flat current — it is a low baseline with periodic wake spikes. The mean is what costs money; the spikes are what tell you your keepalive interval. A plot of current against time over five minutes is one of the better figures this report can contain.

Measure	Method	Report
Idle, connected	INA219, 1 hour at 10 Hz	Mean, and the wake-spike period
Playing, normal volume	INA219 during a real slot	Mean over the slot
Playing, maximum	INA219, tone at full scale	Peak
Daily energy	Computed from the above and the slot schedule	Wh/day and ₹/year

Measure **in the enclosure**, not on the breadboard. Thermal and regulator behaviour differ enough to move these numbers.

RSSI against join success — the interesting analysis

This is the analysis that makes the field trial worth doing, and no published number exists for it in Indian homes.

For every scheduled slot, each box logs: its RSSI at join time, whether it joined within 20 s, join latency, and underruns during playback. Over 10 days and five boxes at five slots a day that is **250 slot records** — small, but enough to see a relationship.

Plot join success against RSSI. The question to answer: **is there a threshold below which slots start being missed, and where is it?** If there is, that number becomes an installation rule — "do not place a box below -X dBm" — which is a genuinely useful engineering output.

Also record, per household: router make and model, approximate distance and walls between box and router, and whether the Wi-Fi password changed during the trial. That last one is not a footnote; it predicts the field-support load for the whole programme.

Telemetry from five boxes

Keep it small. Every box, every slot, one record:

Field	Type	Why
<code>device_id</code>	string	Which box
<code>slot_id</code>	string	Which broadcast
<code>joined</code>	bool	The headline metric
<code>join_ms</code>	int	How fast
<code>rss_i_dbm</code>	int	The independent variable
<code>underruns</code>	int	Audio quality proxy
<code>playout_offset_ms</code>	int	For the spread calculation
<code>volume_pct</code>	int	Disengagement signal
<code>uptime_s</code>	int	Catches silent reboots
<code>heap_free</code>	int	Catches memory leaks

Publish over MQTT or POST to the backend; write every message to SQLite or Postgres. **Nothing about the household, no audio** — the box has no microphone, and saying so in the report is worth a mark.

The 7-day soak

One box, plugged in, on ordinary Wi-Fi, untouched for seven days, playing at the real slot schedule. Started in week 3 so it finishes in week 4 without costing a week.

Pass condition	Value
Missed play commands	0
Unexplained reboots	0
Heap trend over 7 days	Flat — plot it
Recovery after a deliberate Wi-Fi outage	100%, unattended
Audible glitches	≤ 1 per playing hour

The heap plot is the point. A leak that surfaces on day five is precisely the fault class that would otherwise be found by 400 households at once, and a flat line across seven days is a strong result to show an examiner.

Roles, deliverables and assessment

Four roles

Every student owns one subsystem end to end and a share of the report. Rotate pair-work across roles, but keep single ownership — shared ownership of firmware in an 8-week project means nobody owns it.

Role	Owns	Weeks of heaviest load
Embedded / firmware	ESP32 code, state machine, I2S, provisioning portal, telemetry client	1, 4, 5
Hardware / build	Wiring, perfboard, enclosures, power, the five boxes, bench measurement	1, 3, 6
Backend / systems	Icecast, ffmpeg pipeline, scheduler, MQTT or HTTP, database, dashboard	2, 5
Test / analysis	Test plans, latency and power measurement, field data, plots, report editing	1, 7, 8

The test-and-analysis role is not the lightest one and should not be given to whoever is least confident. It carries week 1's experiment, the whole field trial, and every plot in the report — which is most of the marks.

Deliverables

#	Deliverable	Due
D1	Codec and SoC decision note, with the four test results	End week 1
D2	Latency measurement note, with method and percentiles	End week 2
D3	Five working boxes, photographed and labelled	End week 3
D4	Firmware repository, tagged, with a README anyone can build from	End week 5

#	Deliverable	Due
D5	Seven-day unattended scheduler log	End week 5
D6	Field trial dataset, 10 days, 5 boxes, raw	End week 7
D7	Final report	Week 8
D8	5-minute demonstration video	Week 8
D9	BOM with actual prices paid, extrapolated to 500 and 5,000 units	Week 8

Record the demo video. Live demonstrations fail on lab Wi-Fi in front of examiners, reliably and embarrassingly. Have the video, then do the live demo as well.

Report structure

Around 40–60 pages, organised around the two research questions rather than around the weeks:

1. Problem and context — including why no licensed RF path exists, which is the interesting constraint
2. System design and the three endpoint tiers
3. **Experiment 1: codec and SoC.** Method, results, decision, cost consequence
4. Hardware design — schematic, BOM, power budget
5. Firmware design — state machine, provisioning, telemetry
6. Scheduler and unattended operation
7. **Experiment 2: field trial.** RSSI against join success, latency spread, the 7-day soak
8. Cost model — measured BOM extrapolated to volume, against the ₹300 ceiling
9. Limitations and what a production version needs — cite the main guide's Stages 8–10
10. Conclusions

Section 9 is where marks are won cheaply. A team that says clearly what it did **not** do — no certification, no OTA, no PCB, no voice tier — and why, reads as having understood the problem. A team that implies it built a product does not.

Suggested rubric

Criterion	Weight	What full marks looks like
Working demonstrator	25%	Five boxes, five homes, ten days, automatic schedule, no human intervention

Criterion	Weight	What full marks looks like
Experiment 1 — codec	15%	Four tests, real numbers, at a stated RSSI, with a costed decision. A clean negative result scores full marks
Experiment 2 — field data	20%	250 slot records, RSSI-vs-join analysis, a stated threshold or a clear statement that none was found
Engineering rigour	15%	Power measured in the enclosure, latency by a defensible method, heap plotted over 7 days
Cost analysis	10%	Actual prices paid, extrapolated honestly, with the hand-built vs volume gap explained
Report and viva	15%	Limitations stated plainly; every number traceable to a logged measurement

What to do if something fails

A negative result, documented, is worth more than a vague positive one. If Opus does not fit on the C3, that is the answer to the research question and it has a real consequence for the cost model — write it up as a finding, not a failure.

What does lose marks: a demonstrator that only works when a team member is standing next to it, numbers with no stated method, a BOM using list prices rather than what was actually paid, and any claim the data does not support.

Safety, ethics and the traps

Safety

Rule	Why
Never build a mains power supply. Buy BIS-marked 5 V adaptors	Mains work is outside this project's scope and most students' training. The adaptor costs ₹150
No mains wiring inside the enclosure, ever	The box goes into someone's home. Everything inside it runs at 5 V
Solder with ventilation or fume extraction	Flux fumes are an irritant; leaded solder needs handwashing afterwards

Rule	Why
Eye protection when clipping component legs	The commonest actual injury in an electronics lab
Do not test at full volume near your ears	3 W into a 66 mm driver at 10 cm is genuinely loud. Use bench distance
Current-limit the bench supply during bring-up	A miswired amplifier draws a lot of current before it dies
Isolation transformer between audio source and interface	Protects the sound card and removes hum

Ethics and consent for the home trial

Easy to skip, and it should not be. Five boxes go into five real homes and log data.

- ☐ Check whether your institution requires ethics approval for a study in participants' homes. Many do, even for non-medical work, and it can take two weeks — **start this in week 0**
- ☐ Written informed consent from each household: what the device does, what is logged, how long it stays, how to withdraw
- ☐ State plainly, in writing: **the device has no microphone and records no audio**. It is the first thing every participant asks
- ☐ Log no personal data. Device ID, RSSI and join success only — no names or addresses in the database. Keep the mapping on paper
- ☐ A named contact number, and a commitment to remove the box within 24 hours if asked
- ☐ Remove every box at the end. A project that leaves hardware behind in people's homes has not finished

Ten traps that eat a week

In rough order of how often they catch teams:

#	Trap	Avoided by
1	Parts arrive in week 3	Order everything in week 0, in one order, with spares
2	Volume reads garbage whenever Wi-Fi is on	Put the pot on an ADC1 pin. ADC2 is shared with the radio

#	Trap	Avoided by
3	Amplifier is silent and the I2S code gets blamed	The SD pin is shutdown. Tie it high
4	Board stops booting once the front panel is wired	Avoid strapping pins
5	Three weeks lost on Opus-in-Ogg before anything ever made a sound	Get MP3 working first. Then run Opus as the experiment
6	Clicks on loud passages, debugged as a buffer bug	470 μ F bulk capacitance near the amplifier
7	Everything works in the lab, nothing works in a home	Test on a different router from week 2, not week 6
8	Captive portal works on one phone, fails on three	Test on four phones including an old budget Android
9	Field dataset is unusable because firmware changed mid-trial	Freeze firmware before deployment. No exceptions
10	Live demo fails in front of the examiner	Record the video. Then demo live

Traps 2, 3 and 5 are the specific ones. Each is a full week if you meet it unprepared, and eight weeks does not have three spare.

Stretch goals

Only for a team clearly ahead by week 5. **Do not start any of these at the cost of the field trial.**

Stretch	Difficulty	Why it is interesting
OTA update with A/B rollback	High	The real thing. Test only on bench boxes, never on a deployed one
ESP-NOW mesh relay between two boxes	High	Gen 3 of the roadmap. One connected box feeding one unconnected neighbour is a genuinely novel demonstration and the best conference-paper material here
Speech-band DSP on the ESP32	Medium	Spectral tilt and compression tuned for age-related hearing loss. Measurable with a mic and a spectrum plot

Stretch	Difficulty	Why it is interesting
Proper AGC and gate in the encoder	Medium	Replace the crude ffmpeg filter chain with a real one, measured on level consistency across speakers
A second codec comparison, AAC-LC	Low	Widens experiment 1 to three codecs for very little extra work
Designing the PCB in KiCad, without fabricating it	Low	Costs nothing, shows DFM understanding, and is a strong report appendix

What this project is not

Say this in the report, and say it in the viva. The team has built and measured a **demonstrator**, not a product. A production version needs a PCB, WPC and BIS certification, OTA with tested rollback, DLT registration for the voice tier, multi-tenant infrastructure and a 500-household enrolment campaign — Stages 8 through 10 of the main guide, and roughly thirty more weeks.

Knowing exactly where the demonstrator ends and the product begins is the mark of an engineer rather than a student, and it is the cheapest set of marks available in the whole project.